A Research Proposal on

Lightweight Graph Neural Networks for Real-Time QoS-Aware Routing in Software-Defined Networks

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1. Abstract

Ensuring high Quality of Service in large and dynamic computer networks is a growing challenge as data traffic increases. Software-Defined Networking provides centralized and programmable control, making it an ideal solution for enhancing routing efficiency. However, many existing AI-based routing methods, especially those using Deep Reinforcement Learning and Graph Neural Networks, require heavy computation and are difficult to run on smaller or edge-level software-defined Networking controllers in real time.

This research aims to develop a lightweight GNN-based model for quality of service-aware routing in software-defined networks. The proposed approach focuses on reducing network delay, packet loss, and congestion while using fewer computing resources. To achieve this, the project will explore simplified graph neural network architectures and model optimization techniques such as pruning, quantization, and knowledge distillation. We evaluate the performance of the proposed system by simulation-based SDN environments using Mininet and Python tools. The expected outcome of this project would be a lightweight and adaptive routing framework that offers near real-time performance with lower energy use, making it suitable for practical deployment in distributed or edge-based SDN systems.

2. Introduction

Background

Modern networks face growing challenges in providing Quality of Service (QoS) guarantees, such as low latency, minimal packet loss, and high throughput. Traditional routing protocols rely on static or heuristic methods, which cannot adapt effectively to the ever-evolving dynamic traffic conditions. Software-Defined Networking (SDN) offers a promising solution by separating the control and data planes, allowing a centralized and programmable management of network resources [1].

Recently, machine learning-based routing, particularly using Deep Reinforcement Learning (DRL) and Graph Neural Networks (GNNs), has been shown to have a strong potential for adaptive and intelligent routing optimization [2]. GNNs are well-suited for modeling network topologies as graphs, enabling efficient learning of structural and traffic-based relationships. However, most existing GNN-based routing models are computationally intensive, making them impractical for real-time deployment on edge or distributed SDN controllers with limited resources.

Problem Statement

While GNN-based routing methods have been shown to achieve a high accuracy in QoS optimization for SDNs, their heavy computational and memory requirements limit real-world deployment. Edge-level SDN controllers often operate with constrained resources and require fast routing decisions.

Existing DRL and GNN-based models are designed for centralized environments, making them unsuitable for resource-constrained devices. Studies such as [3], [2], and [4] show that deep learning approaches improve network performance but suffer from scalability and latency issues.

This research aims to design a lightweight GNN model that preserves QoS performance while enabling real-time routing on edge-level SDN controllers.

Research Objectives

1. Design and develop a lightweight GNN architecture for real-time QoS-aware routing in SDN.
2. Apply model compression and optimization techniques (pruning, quantization, knowledge distillation).
3. Evaluate the proposed model on edge-level SDN controllers and benchmark against existing algorithms.
4. Analyze trade-offs between routing accuracy, inference latency, and resource consumption.

3. Literature Review

The dynamic and complex nature of modern networks necessitates advanced routing mechanisms, pushing research toward approaches that merge Software-Defined Networking (SDN) with Artificial Intelligence (AI). The core challenge remains in achieving high Quality of Service (QoS) while managing the computational demands of intelligent models.

Early works, such as Kim et al. [5], established the feasibility of Deep Reinforcement Learning (DRL) in SDN by proposing a model that dynamically adjusts link weights to minimize end-to-end delay and packet loss. They employed an offline M/M/1/K queue-based network model to prevent performance degradation during DRL training. Similarly, Song et al. [3] developed a QoS-aware DRL framework using the Deep Deterministic Policy Gradient (DDPG) algorithm and a custom reward function that jointly considered delay, bandwidth, jitter, and packet loss. This centralized approach demonstrated superior efficiency in handling data requests and reducing delay compared to traditional routing methods.

To further enhance adaptability and reliability, later research integrated Graph Neural Networks (GNNs) with DRL. Ampratwum and Nayak [2] introduced a hybrid DRL-GNN model for Wavelength Division Multiplexing (WDM) network restoration, leveraging GNN's topological learning to dynamically select restoration paths and assign wavelengths. Their method improved restoration success rates and resource utilization. Building upon this, He et al. [6] proposed a GNN-based DRL scheme enhanced with causality detection, designed to quantify the causal impact of agent actions on the network environment. This approach achieved superior routing performance and stability under dynamic and noisy conditions.

Finally, Jiang et al. [4] emphasized the importance of GNNs in modeling complex and dynamic network topologies for routing optimization, identifying scalability, real-time deployment, latency, and memory consumption as major bottlenecks.

Research Gap: While DRL and GNN-based routing models demonstrate strong performance and adaptability, their computational intensity limits real-time deployment on resource-constrained SDN edge controllers. This research focuses on developing a lightweight and efficient GNN model suitable for real-time QoS-aware routing in SDN environments.

Table 3.1: Summary of Reviewed Literature

Ref.	Paper Title	Method / Architecture	Key Findings	Core Problem
[5]	*Deep Reinforcement Learning-Based Routing on SDN*	• DRL agent learns optimal link weights • Offline M/M/1/K model	• Reduced delay and packet loss • Outperformed hop-count routing	• High computational cost • Long DRL training time
[3]	*QoS Routing Optimization based on DRL in SDN*	• DDPG-based DRL • Reward balances delay, jitter, loss	• Improved QoS and reduced delay • Validated multi-metric approach	• Poor multi-metric QoS handling
[2]	*Hybrid DRL-GNN for WDM Network Restoration*	• DRL + GNN hybrid model • GNN extracts topology features	• Higher restoration success • Better resource utilization	• Limited adaptability in failures
[6]	*Reinforcement Learning-based SDN Routing with Causality Detection and GNN*	• GNN + causality detection • Quantifies causal influence	• Improved stability and accuracy • Noise-resistant decisions	• DRL instability under noise
[4]	*GNNs for Routing Optimization: Challenges and Opportunities*	• Survey of GNN frameworks	• Identified bottlenecks • Limited real-time deployment	• Scalability and latency issues • High memory usage

4. Methodology

Overview

This research will design, implement, and evaluate a lightweight GNN model for QoS-aware routing in SDN, combining model design, optimization, and experimental validation.

System Design

1. **Network Simulation:** SDN environment setup using Mininet and RYU.
2. **Data Collection:** Capture network metrics under dynamic traffic conditions.
3. **Model Design:** Implement lightweight GNN for routing prediction.
4. **Optimization:** Apply pruning, quantization, and knowledge distillation.
5. **Deployment:** Test on simulated edge SDN controller for latency and memory efficiency.

Evaluation Metrics

- **Network performance:** Average delay, packet loss, throughput.
- **Model efficiency:** Inference latency (ms), memory footprint (MB), CPU utilization (%).
- **Comparative baselines:** Shortest Path Routing, DRL-based routing (DDPG/SAC), and standard GNN models (RouteNet, CeGCN).

Tools and Technologies

Python, PyTorch Geometric (PyG), Mininet, RYU controller, GEANT2 and Internet2 network topology datasets.

5. Expected Outcomes

We expect the following as the expected outcome of the proposed research work:

- The development of a lightweight GNN model for real-time QoS-aware routing.
- Yielding of a lower inference latency and memory usage than existing models.
- Ensuring improved scalability and deployability on edge SDN controllers.
- To ensure the validity and the true performance of the proposed system, we also plan an open-source implementation for reproducibility.

6. Timeline

Table 6.1: Proposed Work Schedule (December 2025 – November 2026)

Task	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Literature Review												
Data Collection & Dataset Preparation												
Implement Baseline GNN & DRL Models												
Develop Lightweight GNN & Apply Compression												
Deploy on Edge-level SDN Testbed & Experiment												
Analyze Results & Optimize Performance												
Prepare Final Report, Paper, & Presentation												

References

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